# Python Notes (Lecture 2)

## Strings

### 1. What is a String?

A string is a data type that stores a **sequence of characters**. Strings are written inside single, double or triple quotes.

```
str1 = "Apna College"
str2 = 'Python'
str3 = """This is a
multi-line string"""

print(str1)
print(str2)
print(str3)
```

Output:

```
Apna College
Python
This is a
multi-line string
```

---

### 2. Basic Operations

#### a) Concatenation

Joining two strings with the `+` operator:

```
first = "hello"
second = "world"

print(first + second)
print(first + " " + second)
```

Output:

```
helloworld
hello world
```

### b) Length of a String

`len()` returns the number of characters in a string (spaces count too):

```
name = "Apna College"
print(len(name))
```

Output:

```
12
```

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## 3. Indexing

Each character in a string has a position called its **index**. Indexing starts from `0`.

| A | p | n | a | _ | C | o | l | l | e | g | e |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |

```
s = "Apna_College"

print(s[0])
print(s[1])
print(s[5])
```

Output:

```
A
p
C
```

Strings are **immutable**: you cannot change a character using its index.

```
s = "Apna_College"
s[0] = "B"      # Error!
```

Output:

```
TypeError: 'str' object does not support item assignment
```

## 4. Slicing

Slicing is used to access a **part** of a string.

```
string[starting_idx : ending_idx]    # ending_idx is NOT included
```

```
s = "ApnaCollege"

print(s[1:4])     # characters at index 1, 2, 3
print(s[:4])      # same as s[0:4]
print(s[4:])      # same as s[4:len(s)]
```

Output:

```
pna
Apna
College
```

- If the starting index is left out, slicing starts from **0**.
- If the ending index is left out, slicing goes till the **end** of the string.

### Negative Index

Negative indexes count from the end of the string. The last character is at index `-1`.

| A | p | p | l | e |
|---|---|---|---|---|
| -5 | -4 | -3 | -2 | -1 |

```
s = "Apple"

print(s[-1])
print(s[-3:-1])   # index -3 and -2
print(s[-3:])
```

Output:

```
e
pl
ple
```

## 5. String Functions

Strings have built-in functions (methods) that we call with a dot:

| Function | What it does |
|---|---|
| `s.endswith("er.")` | Returns `True` if the string ends with the given substring |
| `s.capitalize()` | Capitalizes the first character |
| `s.replace(old, new)` | Replaces all occurrences of old with new |
| `s.find(word)` | Returns the index of the first occurrence (or `-1` if not found) |
| `s.count("am")` | Counts how many times the substring occurs |

```
s = "i am a coder."

print(s.endswith("er."))
print(s.capitalize())
print(s.replace("coder", "student"))
print(s.find("a"))
print(s.count("am"))
```

Output:

```
True
I am a coder.
i am a student.
2
1
```

Note: these functions return a **new** string; the original `s` does not change.

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# Let's Practice : Strings

### Q1. Write a program to input the user's first name and print its length.

```
name = input("Enter your first name: ")
print("Length of your name is", len(name))
```

Output:

```
Enter your first name: Shradha
Length of your name is 7
```

### Q2. Write a program to find the occurrence of '$' in a string.

```python
text = "Hi $ I am $ learning Python $"
print("Occurrences of $ :", text.count("$"))
```

Output:

```
Occurrences of $ : 3
```

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# Conditional Statements

## 1. if-elif-else

Conditional statements let a program make **decisions**: run different code depending on whether a condition is True or False.

```python
if condition1:
    statement1        # runs if condition1 is True
elif condition2:
    statement2        # runs if condition2 is True
else:
    statementN        # runs if none of the above are True
```

- A colon `:` comes after every condition.
- **Indentation** (4 spaces) marks which statements belong to the block.
- Only the **first** True block runs; the rest are skipped.
- `elif` and `else` are optional; you can have many `elif` blocks.

```python
light = "yellow"

if light == "red":
    print("Stop")
elif light == "yellow":
    print("Look")
elif light == "green":
    print("Go")
else:
    print("Light is broken")
```

Output:

```
Look
```

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## 2. Example: Grade Students Based on Marks

| Marks | Grade |
|---|---|
| marks ≥ 90 | A |
| 90 > marks ≥ 80 | B |
| 80 > marks ≥ 70 | C |
| marks < 70 | D |

```python
marks = int(input("Enter marks: "))

if marks >= 90:
    grade = "A"
elif marks >= 80:
    grade = "B"
elif marks >= 70:
    grade = "C"
else:
    grade = "D"

print("Grade:", grade)
```

Output:

```
Enter marks: 84
Grade: B
```

We don't need to write `marks < 90` in the second condition: if marks were 90 or more, the first block would already have run.

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# Let's Practice : Conditionals

## Q1. Write a program to check if a number entered by the user is odd or even.

```python
num = int(input("Enter a number: "))

if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Output:

```
Enter a number: 7
Odd
```

## Q2. Write a program to find the greatest of 3 numbers entered by the user.

```python
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))

if a >= b and a >= c:
    print("Greatest is", a)
elif b >= c:
    print("Greatest is", b)
else:
    print("Greatest is", c)
```

Output:

```
Enter first number: 12
Enter second number: 45
Enter third number: 30
Greatest is 45
```

## Q3. Write a program to check if a number is a multiple of 7 or not.

```python
num = int(input("Enter a number: "))

if num % 7 == 0:
    print(num, "is a multiple of 7")
else:
    print(num, "is not a multiple of 7")
```

Output:

```
Enter a number: 49
49 is a multiple of 7
```